

enVolve: Are You Listening? Inertial Sensing to Monitor the Involvement of Silent Listeners during an Online Interaction

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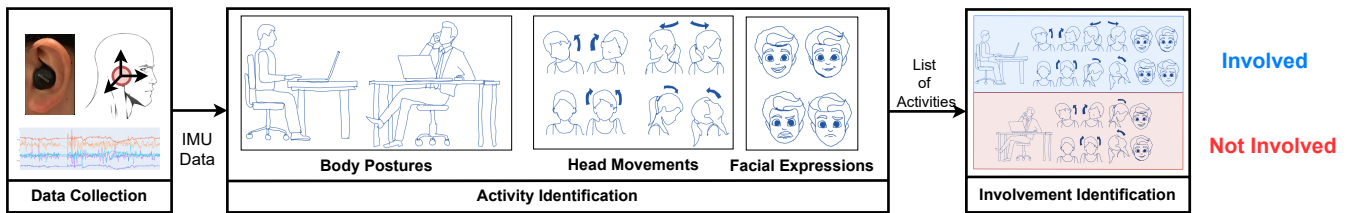


Figure 1: *enVolve* (proposed framework) gathers IMU data using eSense from the listener and recognizes activities during an online class. Then, *enVolve* calculates the level of interest and shows the speaker if the listener is involved or not.

ABSTRACT

Understanding the level of participation for a remote attendee in an online meeting setup could significantly improve the quality of experience for virtual interaction. However, gauging audience involvement over an online meeting becomes particularly challenging when the attendees prefer to turn off the cameras. IMU data have shown promising results in the past to pervasively monitor users' body language, including the determination of various bodily gestures, postures, and facial expressions. This paper demonstrates how earables could help address the stated problem. We provide a motivational study to assess earables for detecting body language corresponding to involved listeners. We further compare it with other sensing modalities like smartwatches and smartphones and accordingly develop a platform called *enVolve*. A lab-scale study with 17 participants demonstrates the efficacy of the proposed system with an average F1 score of more than 80%.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing; • Software and its engineering → Development frameworks and environments; • Computing methodologies → Model development and analysis.

KEYWORDS

online meeting, earables, inertial sensing

ACM Reference Format:

Garvit Chugh, Suchetana Chakraborty, Ravi Bhandari, and Sandip Chakraborty. 2022. *enVolve*: Are You Listening? Inertial Sensing to Monitor the Involvement of Silent Listeners during an Online Interaction. In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct)*, September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3544793.3563419>

1 INTRODUCTION

Several verbal and non-verbal cues, such as gaze, gestures, and posture, in a physical meeting setup help the speaker gauge the audience's involvement continuously. However, the recent trends of online teaching-learning, virtual meetings, and webinars have significantly reshaped the core of human interactions, with the participants not sharing a similar spatio-temporal context on a typical online platform. This results in a significant communication gap regarding low cognitive perception between the speaker and the listeners [3, 15]. This gap is often aggravated due to limited bandwidth and poor network connectivity, thus compelling the users to turn off their webcams while attending an online meeting/lecture.

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UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom

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ACM ISBN 978-1-4503-9423-9/22/09...\$15.00

<https://doi.org/10.1145/3544793.3563419>

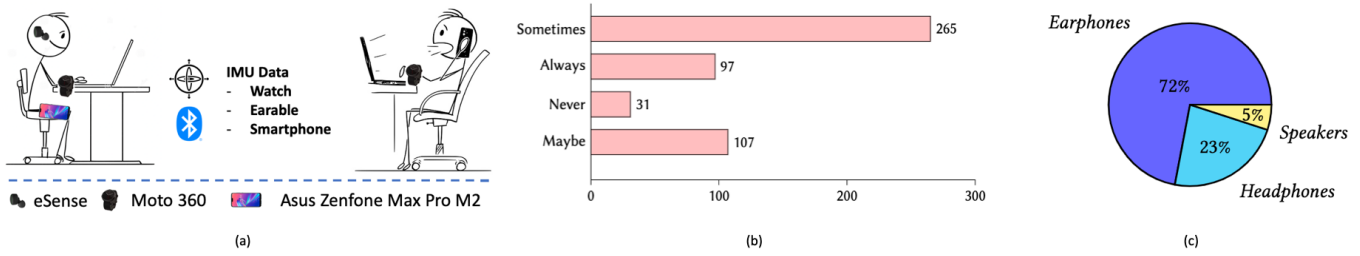


Figure 2: (a) IMU data collection using earable, smartwatch, and smartphone. (b) Survey results of listeners performing movements silently in an online conversation. (c) Survey results of listeners in terms of usage of ear-wears

Moreover, in an online multi-party setup, the audience generally mutes themselves to avoid ambient noise, echo, and interference with the speaker. This limits the usability of audio and video-based approaches for estimating user involvement simply because of the absence of conventional cues such as human gaze, hand gestures, or body posture of the audience. Thus, assessing the listeners' involvement (whether the listener is following or able to follow the talk) in an online meeting setup is particularly challenging for the speaker. While conventionally, based on the visual cues, the speaker could alter the content delivery style, it is difficult to do so in an online setup where the cues are absent. Consequently, the conversation may turn down to be dull and monotonous. Hence, this demands the development of a mechanism that provides real-time feedback to the speaker regarding listeners' involvement in a visually intuitive fashion, thus making online-based content delivery more fruitful to both the speaker and the listeners. In this context, our paper explores using IMU (Inertial Measurement Unit) data obtained from earables to detect different cues generated by the human head and body manifestations. In recent times, earables have drawn considerable interest amongst the research community for analyzing user behavior and detecting visual cues such as human gaze, hand gestures, and other verbal or nonverbal cues. The work of [1, 6, 9, 12] has demonstrated that *eSense* [8] is sensitive enough to identify not just head movements but minute changes in facial muscles as well, which we believe are essential cues for gauging listener involvement. Also, the signals emerging from the earable are less prone to noise due to restrictions on the degrees of freedom of the human head, making the signal much cleaner and usable compared to that obtained from other wearables.

Selecting and identifying features that will help mimic verbal and non-verbal cues is crucial to quantify listener involvement. In this regard, as shown in Fig. 1, our system *enVolve* uses IMU data obtained from *eSense* and detects facial expressions, head movements, and body postures of the listener during online interaction and uses these gestures to understand how well the person is involved in the conversation and synced up with the speaker. The main idea behind this implementation is that in virtual interaction, the silent gestures of the listener act as a reflection of acknowledgment; this acknowledgment could be presented to the speaker as a visual aid to understand the listeners' involvement in the interaction. However, developing a model to infer such acknowledgments is challenging because the IMU data captured from the earables need to be mapped to different gesticulations at different body parts (like facial muscles,

heads, upper body parts, etc.). The inference is needed to be drawn from a collective analysis of such gesticulations. Interestingly, the IMUs from the earables return a complex amalgamation of signals from different body parts. For example, a listener can simultaneously smile and nod their head, and the IMU produces a compound signal from both gesticulations.

In this paper, we develop and implement *enVolve*, a framework for monitoring the involvement of silent listeners in online meetings, on *eSense* IMU-based earbuds. Here, *eSense* is chosen to collect the accelerometer and gyroscope data for the listeners moving their heads in different orientations and paces. For the preliminary study, we used 17 volunteers in the age of 20 – 25 (Female: 12, Male: 5) and recorded 50 – 55 minutes of IMU data for each volunteer while they attended online meetings (online classrooms). From this data, we first explore whether earables are the better choice for inferring the involvement level of silent listeners compared to other sensing modalities like smartwatches and smartphones, and then develop an initial model for the same by exploiting different associated features from the upper body part and facial movements of the meeting participants.

In contrast to the existing works, the contributions of this paper are as follows. (1) We identify features from gestures, such as head movements, facial expressions, and body postures, that mimic verbal and non-verbal cues and demonstrate the efficacy of earables. (2) Our preliminary evaluation reveals that *enVolve* identifies overlapping features with reasonable accuracy (around 80%), typically a complex problem for hand-worn wearables [16, 17]. (3) We present our ongoing attempts toward developing a Google Meet extension that provides the speaker with a visually intuitive interface to gauge user involvement.

2 BIFURCATING COGNITIVE INVOLVEMENTS: DOES EARABLE HELP?

It is not uncommon to find an involved and engaged listener in an online meeting or lecture, to perform affirmative body gestures following the speaker's progress. To mimic the offline classroom process, the listeners' spatio-temporal context must be shared with the speakers. But, *which features of the body movements are helpful to capture the involvement of a listener?* And, *What is the best mode of collecting/capturing these features?*

To answer the first question, we surveyed around 500 students of the *Indian Institute of Technology, Jodhpur* and the *Indian Institute of Technology, Kharagpur* to understand the various features required

for identifying the involvement of a silent listener in an audio-only online interaction. The students who participated in the study belong to undergraduate, postgraduate, and doctorate programs. The participants' ages lie between 18 to 40 years (mean: 20.77, std: 7.49). Based on the survey results, it was observed that (a) the listeners sit in an upright position when they are involved or trying to get involved, (b) there are gesticulations in their facial expressions, and (c) the listeners may acknowledge by nodding their head, or may not nod if they are uninvolved. Thus, head movements, body posture, and facial expressions become important features to capture the spatio-temporal context of the listener. It is to be noted that all of these movements are performed involuntary by most of the participants (Fig. 2(b)). The survey also observed that just a tiny percentage (6%) of the participants believe that they have never indulged in gestures during an online meeting or lecture.

To answer the second question, we observe another outcome of the survey, which reveals that 72% of the participants use earphones while attending online lectures (Fig. 2(c)). This makes IMU-based earables a natural instrumentation choice to understand listener involvement levels. Other methods of collecting IMU data, such as smartphones and smartwatches, have found widespread use for human activity recognition. Notably, most of the features that could be linked with the involvement levels of the listeners are upper body features, which would be difficult to capture using smartphones or smartwatches simply because these are placed in the user's pockets or worn in hand. To test this hypothesis, we instrumented some volunteers with an earable, smartwatch, and smartphone each while attending an online lecture, as depicted in Fig. 2(a). A cursory examination of the IMU data generated from these devices (Fig. 3) shows that earables are more sensitive in capturing the movements emanating in the upper body and hence are ideal to be used in online meetings. We present a further detailed analysis of this in Section 4.

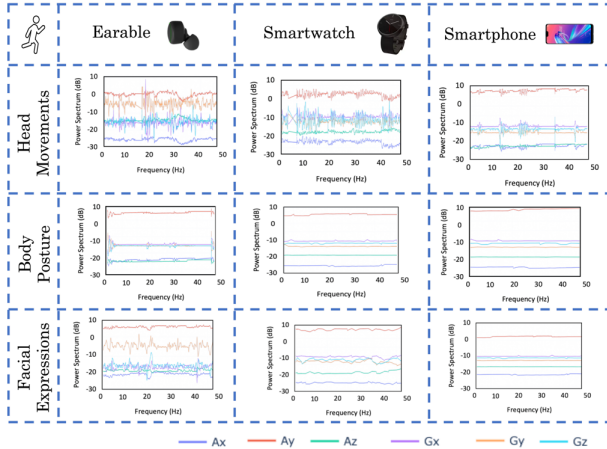


Figure 3: Different IMU Signatures captured from earable, smartwatch, and smartphone for different body movements from one participant at the same timestamp for all devices.

Our findings and prior investigations [6, 8, 9, 12] have indicated that visual clues of human body language, such as an affirmative

Table 1: List of features considered for identifying involvement in *enVolve*

Type of movement	Feature Class
Facial	Inward movement in eyebrows and cheeks
	No movement in eyebrows and cheeks
	Outward movement in eyebrows and cheeks
Body	LaidBack Body Posture
	UprightBody Posture
Head	Head movement in Horizontal Axis
	Head movement in Vertical Axis

head shake (horizontal head movement), perplexed facial expressions (inward cheek movement), or correct body postures (upright), convey evidence of conversational involvement. Combinations of these cues might be interpreted as elements of listener involvement during a conversation. This paper proposes using *head*, *body*, and *facial expressions* to measure listener engagement. Table 1 explains these qualities.

3 ENVOLVE: SYSTEM DESIGN

enVolve comprises two modules: the listener engagement analyzer and the speaker visualizer. First, we present the process for analyzing listener involvement, then how this data is shared with the speaker via a visualizer. Both activities are basic, lightweight, and take very little bandwidth, allowing them to be used online without difficulty, even on a slow machine. Fig. 4 depicts the proposed listener involvement analyzer's processing pipeline.

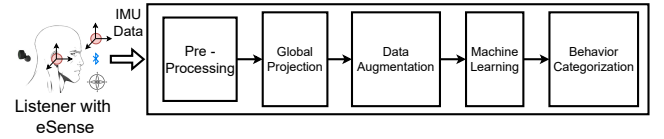


Figure 4: *enVolve* pipeline for involvement identification.

Pre-Processing: *enVolve* gathers 6-axis IMU data from the listener *eSense* [10] and stores it in windows of 10 seconds. Noise and bias must be removed from this data. This phase includes user identification. Integration magnifies noise and bias readings, which impacts the Global Projection (addressed next). Noise is spikes that can be observed without listener movement. This discovery increases the processing window, which improves estimates. Low-frequency signals from head motions are eliminated.

Global Projection: IMU data is saved in its local reference frame; however, movements can happen in such a way that they can cause the default frame of reference to change. As shown in Fig. 1, the head orientation varies according to the users' movement. In order to keep the axes global for all readings, the orientation of the IMU device needs constant rotation. Thus, the data needs to be projected in the global frame where constant changes in the axes can be seen. We use an existing approach like [18] for this purpose.

Data Augmentation: Listeners' features overlap heavily. When a listener acknowledges by nodding, facial expressions help establish/represent participation in the dialogue. Machine learning should separate these actions from the IMU signal. Different data

augmentations/views are created and fed to the models for better learning. We use uniformly random 3D rotation, signal inversion, and order 4 random scrambling for various views. Models are trained with this augmented data. We found that data augmentation improved the models' capacity to discover similarities between unknown and learned data with highly distinguishable feature overlap.

Machine Learning: This phase focuses on the system's machine learning model, which learns to detect the numerous activity features in Table 1 performed by the listener in real-time and shares the prediction with the next step, *Behavior categorization*. A few well-known machine learning models ((a) Convolutional Neural Network (CNN), (b) Long short-term memory (LSTM), (c) Support Vector Machine (SVM), and (d) Decision Tree (DT)) have been picked for proof of concept. Table 2 lists hyperparameters. After *preprocessing*, we generate a three-dimensional array based on the training set's windows, window size, and channels. We use raw signal data for CNN and LSTM (after preprocessing and data augmentation). Others use feature engineering to get statistical and frequency domain feature vectors. Fast Fourier Transform (FFT) is applied to all frequency domain feature vectors. Mean, median, standard deviation, energy measure, skewness, and kurtosis are used to obtain new features. Six FFTs on original 6-axis features and 36 statistical operations on all features create 42 features. Table 1 lists 7 labels for head motions, body postures, and face movements.

Table 2: Hyperparameters of ML Models used

Model	Layers	Filters/Kernel	Filter Shapes	Batch Size	Depth
CNN	3	32	30, 30, 32	16	-
LSTM	4	64	28, 28, 64	16	-
SVM	-	Linear	12, 12	-	-
DT	-	-	-	-	7

Behaviour Categorization: The pipeline's behavior categorization module gets Table 1's collection of listener actions. This module employs a basic heuristic-based technique to link body postures, head motions, and facial expressions, then apply a rule-based approach to determine if a subject is engaging in a meeting's talks during a time window Δ . *enVolve* uses upright body posture as a marker of involvement in its rule-based modeling. We associate body position with head gestures and facial expressions to avoid confusion. This module assumes active listeners have an upright posture, shake their heads to acknowledge, and have involuntary facial emotions, as observed from the survey presented in §2. Further, this module works on the principles of sliding window over Δ , as it processes the model's outcome for every instance of that window and labels either involved or not-involved based on the activities detected; it manages a window of 10 seconds cumulative of all the cases.

The data from the listener involvement analyzer is communicated with the speaker via a visualizer (Fig. 5), which gives feedback on listener engagement. This visualizer works with Chrome and Google Meet. The visualizer has three parts. *First*, it extracts the list of google meet listeners and adds their involvement level. *Second*, it assesses each listener's level and adds a description under their name. *Third*, it calculates the audience's state as a pie chart and delivers analysis to the speaker under the chart. For instance, as shown in Fig. 5, a couple of listeners are lacking behind because

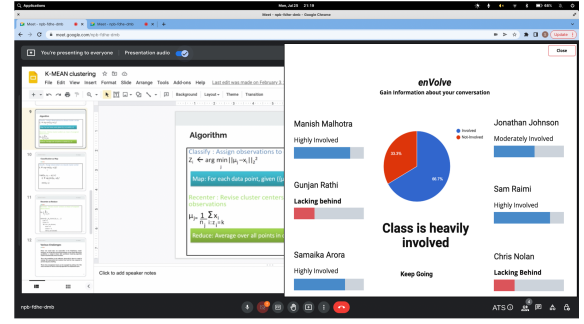


Figure 5: Visualiser providing information about the listeners' involvement level along with overall class status

their engagement level is low; however, at the same time, other students have high engagement levels, creating an overall heavily engaged audience. This visualizer only takes half of the display size to convey the information and does not cause any disruption in the flow of the conversation from the speaker's side.

4 IMPLEMENTATION AND EVALUATION

enVolve is built with compatibility with eSense, an IMU-based earable device. The IMU data, collected at 50 Hz, is recorded through an android application via Bluetooth. The proposed framework *enVolve*'s core is implemented using Tensorflow. The smartphone IMU data is obtained from Asus Zenfone Max Pro M2, and the smartwatch used for the same purpose is Moto 360. Since the data collected is heavily imbalanced, the measuring scale used for this study is the F1 score, which is a better alternative to judge a system's work when dealing with imbalanced data.

4.1 Data Collection

A lab-scale study was conducted at the campus of *Indian Institute of Technology, Jodhpur* to collect the data for the experiment. 17 volunteers in the age range 20 – 25 were recruited to generate the dataset for this study (mean age: 23.52, std age: 1.538, female: 12, male: 5, ethnicity: Indian). Fig. 2(a) shows the setup for volunteers with device placement and the model of the device used for collecting data; these volunteers did not participate in the survey conducted in section 2. For each volunteer, 50 – 55 minutes of accelerometer and gyroscope readings are recorded. To boost the dataset's variance, five volunteers were allowed to surf the Internet and view videos while listening to the speaker; the rest were required to simply keep the browser open to listen to the speaker. These volunteers were videotaped from two angles: (a) front-facing using a laptop webcam and (b) mobile-mounted to capture the whole body. These recordings helped annotators mark the dataset's ground truth. Three non-participating annotators were chosen. They observed participants' body posture, facial expressions, and head movements and marked involved or not involved every second, correlating to accelerometer and gyroscope data in the CSV file. All three annotators marked the final ground truth, but the majority voted. We computed *Cohen's kappa* [14] and obtained a *kappa* value of 0.8 which indicates a good/substantial agreement among the annotators.

Table 3: F1 Score, Precision, Recall scores of testing algorithms for detecting engagement activities

Algorithm	F1 Score	Precision	Recall
Conv-NN	79.64	72.24	80.84
LSTM	77.68	73.65	89.60
SVM	68.82	73.84	84.43
DT	76.67	72.47	79.21

Table 4: Class wise F1 score of CNN for all participants while detecting engagement activities

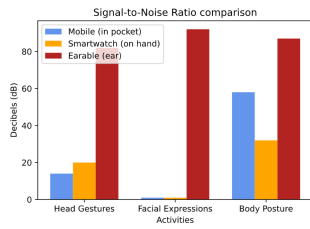
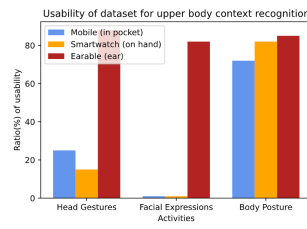
Algorithm	Class Name	F1 Score	Precision	Recall
Conv-NN	Inward Facial Movement	78.13	71.77	80.34
	Outward Facial Movement	80.67	77.15	79.27
	No Facial Movement	76.25	68.87	77.49
	Laid Back Body Posture	77.18	78.35	76.11
	Upright Body Posture	72.97	60.16	70.87
	Horizontal Head Movement	88.89	76.88	91.81
	Vertical Head Movement	83.40	72.54	89.99
Average		79.64	72.24	80.84

Table 5: Class wise F1 score of CNN for all disengaged participants

Algorithm	Class Name	F1 Score	Precision	Recall
Conv-NN	Surfing the internet	70.17	68.49	77.33
	Engaged in other activities	81.23	81.05	82.44
	Average	75.65	74.07	78.44

4.2 Device Type Feasibility

Earables, smartphones, and smartwatches were tested in two ways. *First*, find the gadget with the cleanest raw data. *Second*, Determine the dataset's usability. In these investigations, machine learning algorithms (as mentioned in Section 3) were utilized to identify features and total involvement scores based on IMU data. Our evaluation yielded the following results.

**Figure 6: SNR over data captured over three devices for identifying gestures****Figure 7: Dataset usability for three devices, for identifying gestures**

4.2.1 Signal-to-Noise Ratio (SNR). Here we study the power signals observed and recorded from the three devices. Power spectral density can be generated from an FFT of the IMU data [13]. For the context of activity recognition, we compute the SNR from the power spectral density as $SNR = 20 \log \frac{S}{N}$, where S is the power of the signal for a time window t , and N is the noise power during the previous time window $t - 1$. We divide the captured IMU signals in 10sec windows for this computation. The constant 20 is used

in the above equation, as the signal strength measurements are in watts (power). Fig. 6 shows the results favoring the earable with the highest SNR value amongst all three devices for detecting target activity features like the change in body posture, head gestures, and facial expressions.

4.2.2 Usability of dataset recorded from multiple devices. Here we study the usability of the dataset for upper body context recognition recorded from the three devices. We collected data from the three modalities from 5 different participants and ran the classifiers for detecting various body movements shown in Table 1. The ratio was taken from the collected dataset to ascertain how many times we could identify the different activities correctly and shown in terms of % in Fig. 7. With the same parameters as shown in Table 2, CNN and SVM were used to classify different activities from the data collected from all three devices. It can be seen from Fig. 7 that smartphones and smartwatches are not quite sensitive to recording the minute movements of facial muscles and seem to lose signal power while reaching head movements. A similar observation can be made from Fig. 3, that earables provide much cleaner IMU signatures even for sensitive muscle movements. One interesting observation can be seen that all three devices are more than capable of identifying the body postures with high accuracy.

4.3 Performance Benchmarking of enVolve

enVolve was tested on two parameters: correctly identifying listeners' engagement level during online interaction using CNN, SVM, LSTM, and DT (described in Section 3) and measuring the framework's runtime resource usage. The benchmarking compares the system's memory and bandwidth usage with and without the suggested framework. Benchmarking shows the system's real-world performance and usefulness.

4.3.1 Model Performance. Table 3 summarizes the performance of different machine learning models in correctly inferring the activities mentioned in Table 1, performed by the user during the interaction. It can also be seen from the results and Fig. ?? that our proposed framework can correctly identify multiple overlapping features with a good F1 score of nearly 80%, along with a high recall value. The results of the best performing model, i.e., CNN, is expanded further, *firstly*, in Table 4, which exhibits class-wise F1 score of all the classes mentioned in Table 1; *secondly*, in Table 5, which exhibits class-wise F1 score of all the activities resembling disengagement. Based on the predictions made by the machine learning models, the behavior categorization module of the pipeline (Fig. 4) can correlate these predictions with the state of the listener.

4.3.2 Benchmarking of enVolve. We benchmark the memory and bandwidth consumption by enVolve to test its usability over hand-held devices. This benchmarking has been done for over 5 subjects in a live classroom setup. The online interaction was completely audio mode from the listeners' side; however, the speaker shared a presentation while speaking. The online classes lasted for 50 minutes, during which the bandwidth and memory consumption changed due to the speaker's presentation sharing, which added extra load on the browser to use more memory. The framework's use hasn't greatly increased resource consumption, according to this evaluation. After a few minutes of a minor increase from 8% to

27% RAM utilization, it achieved saturation. Overall Memory usage increased by 10% for 50 minutes. Bandwidth use is similar, with an initial increment of 26%, saturation was achieved after a few minutes, and an overall 13% climb for 50 minutes. This benchmarking provides enough evidence that the system in its current state is useful and provides good foundational support for scalability.

5 RELATED WORK

In recent times, inertial sensing through earables has gained prominence in determining human activities and behavior [6, 10, 19]. The IMU data generated from earables such as eSense devices have exhibited promising results in characterizing gestures like head nodding, head shaking, chewing, speaking, etc. The user study performed by Choi *et al.* [4], conducted with the help of earable devices and wristbands, revealed head nodding and hand gestures can be utilized to determine a meeting's success. According to [4], a meeting has three stages: initiation, group discussion, and turning points. Hand and head gesture cues were calculated for these phases with 79% accuracy. However, unlike *enVolve*, their study was not concerned about providing feedback to the speaker. Also, the study participants were involved in marking the ground truth, which can break the flow of the audience to perceive the lecture as they are focusing on filling the survey during the lecture; hence, an automated method is required. Another similar work [11] has attempted to use earables to detect the participants' head movements and body posture in an online meeting. They, however, have not quantified the involvement level of the participants. Moreover, in this work, the participants were also involved in marking the ground truth; thus, the model may introduce subjectiveness in the evaluation.

Gashi *et al.* [5] have used earables to learn the movements of facial muscles to classify gestures such as smile, yawn, talk, confusion, etc., with an F1 score of 84%. However, independent classification of head gesticulations, body posture, and facial expressions will not address the problem of determining the level of involvement of passive listeners in an online meeting. Although some of the past efforts [11, 20] have attempted using inertial sensing to gauge the involvement of the meeting attendees, they lack in providing continuous feedback about the listeners to the speaker, which questions the effectiveness of the solutions in a real-time scenario.

Other approaches such as [2] and [7] have tried to provide real-time feedback and assistance by analyzing user behavior. However, their methodology uses audio and video data, which requires additional computation and resources. Our implementation works as a low-cost yet effective tool to analyze user behavior without significantly adding or increasing the load on the system. To the best of our knowledge, none of the existing works till now have tried to correlate the speaker-listener using earables during an online meeting in the absence of any *acoustic-visual cues*.

6 DISCUSSION AND CONCLUSION

enVolve demonstrates promising results in passively estimating the involvement level of a silent listener in an online meeting using the IMU data collected from eSense earable. Based on the positive results from this study, several other objectives have risen, such as an end-to-end system based on earables that can provide individual and global learning, and the attention level of a specific user, while

improving the current visualizer to add more features. Another potential objective is to detect or derive different characteristics of individual users, such as their emotional well-being, using the same data. We plan to improve the current accuracy of the models and implement a novel technique based on the learning objectives of the models used in this study.

This investigation yielded relevant and fascinating findings. Cognitive expressions vary by person. Such signals can differ even for a single human. In such instances, it's difficult for machines to learn user context and personalize services. We need a gesture-invariant technique to model the system to eliminate subjective bias from individuals. Continuous and pervasive monitoring of behavioral gestures and psychophysiological markers can give individualized intelligent services on demand.

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